THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 18 u	2×1 + 16
2	**(d)** 36.5 u	1 + 35.5
3	**(a)** 4.44 g	6.92 − (1.76 + 0.72) = 4.44 g
4	**(c)** 2 g and 16 g	1 : 8 by mass
5	**(b)** $CaCl_2$	Ca^{2+} and 2 Cl^-
6	**(b)** carbon disulfide	prefixes show the number of atoms; second element ends in -ide
7	**(b)** Antoine Lavoisier	father of modern chemistry
8	**(c)** covalent bond	shared electron pair
9	**(c)** Cl^- ion	chlorine gains one electron
10	**(a)** in molten state and in aqueous solution	ions are free to move

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	In a chemical reaction matter can neither be created nor destroyed; total mass of reactants equals total mass of products (2).	2
12	In the open set-up the CO_2 produced escaped; in the closed set-up it stayed inside, so the total mass was unchanged (2).	2
13	Reactants $= 6.92$ g; products $= 1.76 + 0.72 + 4.44 = 6.92$ g; equal, hence verified (1+1).	1+1
14	Carbon 6 g needs 16 g oxygen, so $6 + 16 = 22$ g of CO_2 (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Weigh both solutions together before mixing (1); on mixing a white precipitate of barium sulfate forms with sodium chloride (1); the total mass remains unchanged — the law is verified (1).	1+1+1
16	Total reactants $= 6.92$ g (1); mass of products must equal this (1); $CaCl_2 = 6.92 - (1.76 + 0.72) = 4.44$ g (1).	1+1+1
17	No (1). The gaseous products (CO_2 and water vapour) escape into the air (1), so a closed system would show the same total mass (1).	1+1+1
	OR: 12 g C → 44 g CO_2; 2.4 g C → $\frac{44}{12} \times 2.4 = 8.8$ g (2); Law of Conservation of Mass / Constant Proportions (1).	2+1
18	Cl $= \frac{35.5}{23} \times 46 = 71$ g (2); NaCl $= 46 + 71 = 117$ g (1).	2+1
	OR: S : O = 40 : 60 = 2 : 3 (1); oxygen $= \frac{3}{2} \times 20 = 30$ g (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mass of reactants equals mass of products (1). (ii) Weigh flasks containing the two solutions, mix them to get a white precipitate of barium sulfate, reweigh; the masses are equal (3). (iii) Escaping gas would carry away mass (1).	1+3+1
	OR: (i) $6.92 - 2.48 = 4.44$ g (2). (ii) Law of Conservation of Mass (1). (iii) Final mass would appear less (1). (iv) CO_2 would escape (1).	2+1+1+1
20	(i) A compound always contains the same elements in a fixed mass ratio; Proust (1). (ii) Water is a compound; its composition does not depend on the source (1). (iii) H : O = 1 : 8 → H 4 g, O 32 g (2). (iv) No — mixtures can have any ratio (1).	1+1+2+1
	OR: (i) 71 g (2). (ii) 117 g (1). (iii) 46 g (1). (iv) Law of Constant Proportions (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 44 u	12 + 2×16
2	**(c)** 58.5 u	23 + 35.5
3	**(a)** 44 g	mass is conserved
4	**(d)** 71 g	$\frac{35.5}{23} \times 46 = 71$
5	**(a)** Na_2SO_4	2 Na^+ and SO_4^{2-}
6	**(d)** dinitrogen tetroxide	prefixes show the number of atoms; second element ends in -ide
7	**(d)** Joseph Proust	fixed mass ratio in a compound
8	**(d)** a double bond	two shared pairs
9	**(c)** ionic bond	electrostatic attraction
10	**(c)** low melting and boiling points	weak intermolecular forces

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A compound always contains the same elements in a fixed ratio by mass irrespective of the source (1); water: H : O = 1 : 8 (1).	1+1
12	H : O = 1 : 8, so 1 g H and 8 g O (2).	1+1
13	In a compound elements are chemically combined in fixed ratio; in a mixture the components can be mixed in any ratio (2).	2
14	All matter is made of atoms; atoms are indivisible and cannot be created or destroyed; atoms of an element are identical; atoms combine in simple whole-number ratios (any two) (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{4}{3} \times 9 = 12$ g (3).	1+1+1
16	$8 : 2 = 4 : 1$, the same ratio (2). So yes — the compound has a fixed ratio irrespective of how it is prepared (1).	2+1
17	All matter is made of atoms (1); atoms are indivisible and are neither created nor destroyed in reactions (1); atoms of an element are identical, of different elements different; they combine in simple whole-number ratios (1)	1+1+1
	OR: (i) Atoms are neither created nor destroyed, only rearranged, so mass is conserved (1.5). (ii) Atoms combine in fixed simple ratios, giving fixed mass ratios (1.5).	1½+1½
18	Cl (2,8,7) needs one electron; each atom shares one electron, forming one shared pair (1+1); Cl—Cl single bond, each atom now has 8 electrons (1).	1+1+1
	OR: O (2,6) needs two electrons; each shares two electrons (2); two shared pairs give a double bond O=O (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Atoms; indivisible; identical for an element; different for different elements; simple whole-number ratios; constant relative number (3). (ii) Atoms are conserved ⇒ mass conserved; simple ratios ⇒ fixed ratio by mass (1). (iii) Atoms are divisible into subatomic particles (electrons, protons, neutrons) (1).	3+1+1
	OR: (i) A bond formed by sharing electron pairs (1). (ii) Each H shares 1 e^- forming H—H; each Cl shares 1 e^- forming Cl—Cl; H and Cl share 1 pair forming H—Cl (3). (iii) O_2: two pairs (double bond); N_2: three pairs (triple bond) (1).	1+3+1
20	(i) H: 1; O: 2,6 (1). (ii) O shares one electron with each of two H atoms giving two single bonds; each H gets duplet and O gets octet (2). (iii) A compound has properties different from its elements (1). (iv) 1 : 8 (1).	1+2+1+1
	OR: (i) Na loses one electron, Cl gains it; the ions are held by electrostatic attraction (2). (ii) Na^+ (2,8), Cl^- (2,8,8) (1). (iii) MgO: Mg (2,8,2) loses 2 e^- to O (2,6) forming Mg^{2+} and O^{2-} (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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